

PrP-V127 as a Modular Antiprion Platform: Therapeutic Vectors, Tissue-Scale Transport Design, and Regulatory Precedents for Creutzfeldt-Jakob Disease

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Draft for bioRxiv deposition — establishes priority of the secretor/mRNA vector designs and design rules described herein.

Abstract

Prion diseases are fatal with no approved therapy. The human PrP variant G127V, selected during the kuru epidemic, confers complete prion resistance and acts as a potent dominant-negative inhibitor of wild-type propagation — including as a GPI-anchorless, secreted protein in trans (in vitro 2026; AAV proof-of-concept in rodents 2026). We present a structured preclinical review and research program that converts this molecular asset into a deployable therapy for CJD. (1) **Vectors:** we audit three delivery modalities for the anchorless V127 agent — CRISPR-edited neural progenitor secretory grafts, recombinant protein, and intrathecal LNP-mRNA — with pre-registered go/no-go gates in sCJD-infected human cerebral organoids (platform validated: infection 2019; PPS drug screening 2021; NPC seeding 2023; V127 in any form never tested). (2) **Transport design:** a self-tested advection-diffusion-reaction solver (mass conservation 100%; Thiele-length error 0.5%) yields three falsifiable design rules — containment-ring node spacing 8-12 mm; hydrogel carrier mesh $\xi \geq 5\times$ the protein radius (hyaluronic acid 1-2% approved, >5% retains the secretome); mRNA redosing ≤ 7 days; and a containment shell of 4.2-9.5 mm per deposit depending on the capping/replication ratio, a parameter directly measurable in the proposed organoid assay. (3) **Regulatory precedents:** by analogy to sister proteinopathies, no pillar requires an unprecedented regulatory category — chronic intrathecal redosing (SMA/nusinersen), accelerated approval via biomarker in lethal genetic neurodegeneration (SOD1-ALS/tofersen, NfL), cell grafts in human brain (Parkinson 2026), dominant stabilization of a misfolding-prone native protein (TTR/tafamidis), and substrate silencing (ASO/RNAi) are all approved categories whose conjunction this program requests. We conclude that anchorless-V127 antiprion therapy is publishable now as a program and testable within one organoid cycle (~10 months), with realistic endpoints of significant slowing (structured estimate 30-45%) rather than cure.

1. Background and scope

CJD is uniformly fatal within months (sporadic) although genetic carriers (E200K, D178N, P102L) afford a presymptomatic window of years — the population where autologous, time-permissive protocols are feasible; a first E200K kindred in Brazil was reported in 2007. This manuscript integrates: a 41-reference audited review (including correction of recurring citation errors in the field), refutation-driven solution branches, quantitative transport engineering, and a regulatory analogy map. It is the program-level companion to the organoid gate protocol (G0).

2. The molecular asset and its three vectors

G127V: complete resistance when biallelic (heterozygotes remain infectable by vCJD — hence biallelic editing or anchorless delivery is mandatory); structural basis twofold (conformational restriction of the pre- β sheet, dimer stabilization via intermolecular H-bonds, $\beta 2$ - $\alpha 2$ loop dynamics). Dominant-negative effect: dose-dependent in cis; **and in trans as recombinant anchorless protein (2026), with protection persisting after transgene cessation** — the property that makes transient vectors (mRNA) viable and reduces dependence on permanent "factory" designs. In vivo proof-of-concept: systemic AAV delivering anchorless V127 extended survival ~50 days in a rodent model.

Vectors: **(a) Secretory neural progenitor grafts** (CRISPR biallelic V127 + Δ GPI secretion; optional iPSC-microglia co-graft for phagocytic support; hypomune HLA-KO+CD47 platform for off-the-shelf banking); **(b) recombinant anchorless V127 protein** (direct chemical comparability with the 2026 in vitro data); **(c) intrathecal LNP-mRNA** (expression in ~30% neurons / 38% astrocytes after a single intrathecal dose in

rodents; manufacturing in days — the only vector compatible with compassionate use in sporadic CJD timelines).

3. The organoid gate (G0): pre-registered falsifiability

Eight arms, n=8 organoids/arm, ~10-month cycle, BSL-3: A1 mock; A2 disease control; A3 unedited-cell control; A4 membrane-V127 (cis only); **A5 secretory graft (thesis)**; A6 recombinant protein; **A7 LNP-mRNA (transient)**; A8 pentosan polysulfate as published positive control benchmark. Primary readout: the proximal($\leq 1\text{mm}$)/distal($\geq 3\text{mm}$) PrP-res gradient — the spatial signature of trans dominant-negative action, with expected 4-6 mm halo scale from transport modeling. Pre-registered kill-switches include the pivot to acellular vectors if $A6 \approx A5$.

4. Transport engineering (WS-7): three design rules

An ADR solver on heterogeneous porous medium (ECS $\alpha=0.20$, $\lambda=1.8$; $D_{\text{eff}} \approx 3.9 \times 10^{-11} \text{ m}^2/\text{s}$; first-order consumption parametrized 10^{-6} - 10^{-5} s^{-1}): **Rule 1** containment-ring node spacing 8-12 mm (protection radius $r_{10\%} \approx 4\text{-}6 \text{ mm}$ per node); **Rule 2** hydrogel mesh $\xi \geq 5 \times r_p$ — HA 1-2% releases the secreted agent ($D_{\text{gel}}/D_0 \geq 0.7$), >5% retains it; **Rule 3** mRNA redosing interval ≤ 7 days keeps the inter-pulse trough $\geq 30\text{-}56\%$ of peak. Wave-vs-shield analysis: a 1 mm deposit contains an advancing front within a 4.2-9.5 mm shell for capping/replication ratios 0.1-0.01 — overlapping neighbor shells, and the ratio itself is what G0 measures.

5. Regulatory analogy map: nothing unprecedented is being asked

Program pillar	Approved precedent in sister disease
Chronic intrathecal redosing	SMA — nusinersen (2016), years of safety in children
Accelerated approval via biomarker in lethal genetic neurodegeneration	SOD1-ALS — tofersen (FDA 2023, NfL endpoint; sponsor also runs the prion ASO program)
Cell grafts in human brain	Parkinson — dopaminergic progenitor trial (Lund, 2026)
Cell-as-protein-factory	Parkinson — GDNF-delivering grafts (MJFF-funded)
Dominant stabilization of a misfolding-prone native protein	ATTR amyloidosis — tafamidis
Substrate lowering	ATTR/SMA — ASO/RNAi class

The program therefore requests the **conjunction of approved categories**, not a new category — the central de-risking argument for sponsors and agencies.

6. Ethics and limitations

Population: presymptomatic genetic carriers (autologous, time-permissive) with compassionate sporadic use only via the acellular mRNA vector. Falsified or retracted evidence is excluded by rule (e.g., minocycline/FK506 hamster study, retracted 2020; minocycline also confounds NfL interpretation). Limitations: transport parameters are parametric sweeps pending G0 measurement; organoid readouts proxy, not predict, clinical slowing; structured probability estimates (significant slowing 30-45%) are not trial data.

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